

## A generalized Maxwell-Wagner theory for membrane polarization in shaly sands

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### ABSTRACT

The effects of charged clay platelets on the frequency dependent electrical properties of shaly materials are analyzed using simplified models for the membrane polarization around charged spheres immersed in electrolytic solutions, under a thin double layer approximation. The polarization is defined through two possible mechanisms: (1) a surface conductivity related with a modified Stern double layer model (S-model) according to Schurr-Schwarz theory; (2) a coupled electro-diffusional mechanism occurring in a Guoy-Chapman double layer using Fixman's approach (D-model). By comparing the electric potential in such microscopic models with the external potentials derived for the equivalent homogeneous sphere using a Maxwell-Wagner approach, we obtain the total current conductivity functions for these two models. The theory, therefore, provides explicit expressions relating the total conductivity functions to the model parameters.

The behavior of the S-model is described by a complex conductivity exhibiting a simple Debye characteristic. In the D-model, both the conductivity and the dielectric permittivity are given as complex properties, showing similar but much wider dispersion than that of a Debye substance. Our representation of the grains and their associated ionic double layers by an equivalent sphere with effective properties allows us to extend our results to simulate rocks containing clays. This is accomplished using generalized mixture equations written in terms of the total conductivity functions. The frequency behavior of both models are compared and their fit to experimental data on clay-water systems and shaly materials suggest that the D-model is more appropriate for representing the dielectric behavior of clay bearing rocks. The theory can be adapted to estimate the clay parameters of a shaly sandstone using electromagnetic borehole measurements.

### INTRODUCTION

Systematic experimental measurements of the electrical properties of shaly materials have shown that both their conductivity ( $\sigma$ ) and permittivity ( $\epsilon$ ) are dispersive. Although individually for water and most silicate minerals these properties are almost independent of frequency ( $\omega$ ), the composite properties vary appreciably with  $\omega$  (Olhoeft, 1985). In the frequency range from 1 to  $10^6$  Hz, the effective conductivity is a slowly increasing function of  $\omega$ . The effective dielectric constant  $k(k = \epsilon/\epsilon_0$ , where  $\epsilon_0$  is the permittivity of vacuum), on the other hand, is much more strongly dependent on  $\omega$ , starting with values above  $10^3$  at a few Hertz and decreasing to an asymptotic value of less than 80 at high frequencies (Fuller and Ward, 1970; Sen et al., 1984; Vinegar and Waxman, 1984). Furthermore,  $\sigma$  and  $k$  are

also dependent upon the salinity of the electrolyte (Olhoeft, 1985; Johnson and Sen, 1988; Sen et al., 1988).

These reported characteristics have been interpreted as being caused by geometric or textural heterogeneities of the rock system (Sen, 1981; Sen et al., 1984; Schwartz et al., 1989; Knight and Endres, 1990), or were related to electrical and electrochemical processes developed at the interfaces between clay minerals and electrolytic solutions (Marshall and Madden, 1959; Dias, 1972; Sen et al., 1984). Interactions between charged clays and aqueous electrolytes give rise to an ionic double layer around these particles. Polarization of such a layer by an applied electric field, has been pointed out as the main mechanism for the anomalous behavior observed in rocks, soils, and other biphasic systems of charged particles (Dukhin and Shilov, 1974; Fixman, 1980; Chew and Sen, 1982; Grosse and Foster, 1987).

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Several previous models have attempted to explain the frequency dependent electric behavior of rocks on the basis of a Maxwell-Wagner polarization mechanism (Wobschall, 1977; Lysne, 1983; Feng and Sen, 1985). The rock grains were assumed to have fixed (frequency independent) dielectric properties as was the saturating fluid. Large inconsistencies between such models and experimental data are observed. These models are not adequate in simulating shaly sand measurements mainly because they do not properly account for the electromigration and diffusion of ions which occur at the particle-fluid interface (Lyklema et al., 1983).

In a previous paper (Lima and Sharma, 1990), we developed a practical procedure to describe the low-frequency conductivity of a shaly sandstone by using simple granular models where the clays occur either as continuous coatings over the sand grains or as isolated conducting particles. The essence of that method was a procedure to express the surface conductivity of clay particles by an equivalent volume conductivity. This procedure gave excellent agreement with laboratory and well log resistivity data.

The objective of the present work is to use a similar approach and apply it in the frequency domain. We choose to represent the clay effects by total current conductivity functions as proposed by Dias (1972). These functions are derivable from solutions for the electric potentials around charged grains and their counterion atmosphere (Schurr, 1964; Fixman, 1980), in conjunction with a generalized Maxwell-Wagner treatment of the suspended sphere problem. In the frequency range considered here (1 Hz to 10 MHz), the diffusion coupling mechanism is the most important effect in the electrical dispersion of shaly rocks. Other coupling processes, such as the electro-osmotic and electrokinetic effects that may have significant contributions if the sphere were moving, are not considered here.

To simulate shales and shaly sands, we construct composite materials using generalized effective medium theories (Hanai, 1968; Sen et al., 1981; Feng and Sen, 1985). For each particular model, the total current conductivity functions are decomposed into the corresponding complex physical properties and the effective experimental parameters that describe the behavior of the binary system. Through this procedure it is possible to represent the response of a wet charged sphere aggregate in terms of equivalent homogeneous complex electrical properties. In this way, we present an integrated approach to describe the dispersive behavior of both the conductivity and permittivity of shaly rocks.

The analytical results can be used to describe the phenomenon of frequency dependent charge polarization in clay rich water-saturated rocks and to explain the observed frequency and salinity dependence of the electrical behavior of these materials. They also serve as a fundamental basis for explaining and designing simple equivalent circuit analogs for rocks. Furthermore, the low-frequency limits of these functions support our previous results for dc currents (Lima and Sharma, 1990).

Electromagnetic logging tools now in use, operate within a frequency range where the effects of distributed clays in rocks are most prominent. By having an appropriate physical model for such rocks, one can extract from well log measurements useful information regarding the clay type, its proportion, and distribution through a porous reservoir.

## BASIC THEORY

The macroscopic electrical behavior of conducting dielectrics subjected to harmonic sinusoidal fields is described by Maxwell's equations and certain constitutive relationships. For a time dependence  $\exp(-i\omega t)$ , the constitutive dielectric equation, Ohm's law and Maxwell's equation for the total current density ( $\mathbf{J}_T$ ) in a medium can be combined to obtain,

$$\mathbf{J}_T = (\sigma^* - i\omega\epsilon^*)\mathbf{E}, \quad (1)$$

where  $\sigma^*$  and  $\epsilon^*$  are, in general, complex scalars dependent upon frequency (Fuller and Ward, 1970; Dias, 1972).  $\mathbf{J}_T$  represents the sum of two current densities: one due to the drift of free charges under the action of the applied electric field and of local physico-chemical potentials; the other is related to the oscillations in the distribution of bound charges. In the frequency domain,  $\mathbf{E}$  is the applied electric field.

Writing  $\sigma^* = \sigma' + i\sigma''$  and  $\epsilon^* = \epsilon' + i\epsilon''$ , in terms of their real and imaginary parts, equation (1) is rewritten as

$$\mathbf{J}_T = (\sigma_{ef} - i\omega\epsilon_{ef})\mathbf{E} = \bar{\sigma}^*\mathbf{E}, \quad (2)$$

where  $\sigma_{ef} = \sigma' + \omega\epsilon''$  and  $\epsilon_{ef} = \epsilon' - \sigma''/\omega$  are real functions of frequency that constitute the experimentally measured rock parameters, and  $\bar{\sigma}^*$  represents the total current conductivity function proposed by Dias (1972). Thus, the real effective conductivity of a substance includes contributions from free charge conduction as well as some

## LIST OF SYMBOLS

|                  |                                                            |
|------------------|------------------------------------------------------------|
| $a$              | = Particle radius                                          |
| $D$              | = Ionic diffusion coefficient                              |
| $\mathbf{E}$     | = Electric field                                           |
| $E_o$            | = Electric field intensity                                 |
| $e_o$            | = Electronic charge = $1.6022 \times 10^{-19}$ C           |
| $\mathbf{J}$     | = Current density                                          |
| $k_B$            | = Boltzmann constant = $1.38 \times 10^{-23}$ J/K          |
| $T$              | = Absolute temperature                                     |
| $p$              | = Volume clay fraction                                     |
| $\Psi$           | = Electric potential                                       |
| $\epsilon^*$     | = Dielectric permittivity                                  |
| $\sigma^*$       | = Electric conductivity                                    |
| $\epsilon_o$     | = Permittivity of free space = $8.854 \times 10^{-12}$ F/m |
| $\bar{\sigma}^*$ | = Total current conductivity                               |
| $\mu$            | = Ionic mobility                                           |
| $\lambda$        | = Surface conductivity                                     |
| $\rho$           | = Charge density                                           |
| $\tau$           | = Relaxation time                                          |
| $\beta$          | = Surface counterion density                               |
| $\phi$           | = Porosity                                                 |
| $\kappa$         | = Debye screening distance                                 |

Single and double primes denote real and imaginary parts of complex functions. Subscripts  $w$ ,  $c$ , denote pore water and clay, respectively. Additional subscripts  $ef$ ,  $0$ , and  $\infty$  denote the effective bulk, low-frequency and high-frequency properties, respectively.

effects due to polarization mechanisms. Conversely, the effective permittivity has contributions from both bound and mobile charge carriers.

Using equation (2) and applying the classical Maxwell-Wagner treatment, it can be shown that the external potential distribution ( $\Psi_w$ ) around a spherical particle of arbitrary total current conductivity  $\bar{\sigma}_c^*$  suspended in a medium with total current conductivity  $\bar{\sigma}_w^*$  and subject to an alternating electric field, can be written as

$$\Psi_w = \left[ -r + \frac{(\bar{\sigma}_c^* - \bar{\sigma}_w^*)a^3}{(\bar{\sigma}_c^* + 2\bar{\sigma}_w^*)r^2} \right] E_o \cos \theta, \quad (3)$$

where  $E_o$  is the electric field intensity,  $a$  the particle radius, and  $r$  and  $\theta$  the spherical coordinates. It is important to note that in the present case,  $\bar{\sigma}_c^*$  is the total current conductivity function of the solid grain and its associated electrical double layer. It incorporates within it, double layer polarization effects. Equation (3) is employed in the following subsections to compute the total current conductivity function of a suspended charged sphere by solving for the ionic fluxes in the double layer. Two limiting conditions are considered: (a) bound layer polarization; and (b) diffuse layer polarization.

#### Bound layer polarization

A simplified theory for the dielectric dispersion of a system of charged particles based on a surface conductivity mechanism has been considered by O'Konski (1960), Schwarz (1962), and Schurr (1964).

O'Konski (1960) introduced an integral surface conductivity operating at the interface between two different media to represent its excess electrical conduction. His boundary value problem for spheres within electrolytes was solved by allowing a free exchange of charges between the surface layer and the electrolyte, but neglecting any ionic diffusion along this layer. His constant conductivity is satisfactory at high frequencies, but it is not adequate in representing the dielectric behavior of charged systems below the characteristic frequency of the Maxwell-Wagner effect (Schurr, 1964; Lyklema et al., 1983).

Schwarz's approach (Schwarz, 1962) incorporates a strong capacitive effect by just considering the diffusive fluxes of counterions in a spherical ionic atmosphere. In fact, his double layer was treated as a thin layer of tightly bound counterions so as to prevent their exchange with ions in the bulk electrolyte. These bound ions are, however, free to move tangentially along the surface of the sphere. Schwarz's theory agrees satisfactorily with experimental effective permittivity data on polystyrene spheres even for frequencies below 10 Hz (Chew and Sen, 1982). However, his neglect of continuous charge transport across the double layer leads to an inconsistent dc result of zero equivalent conductivity for a charged insulating particle having a counterion layer.

In Schurr's theory (Schurr, 1964) the ionic atmosphere around a charged sphere immersed in an aqueous electrolyte is divided into two parts, one represented as a thin layer of fixed counterions with density  $\rho_b$ , and another as a diffuse layer with ionic density  $\rho_d$ . The composite double layer is thin in comparison to the particle radius and has zero total net charge. The following conditions were additionally introduced: (1) the fixed or Stern layer charges are only free to

move tangentially; the electromigration currents are relaxed by opposing currents due to ionic concentration gradients; (2) the diffuse layer charges exchange readily with the bulk electrolyte; thus, there is no polarization in the diffuse atmosphere; (3) no ions are exchanged between the fixed and the diffuse layers even in the presence of an alternating field. Under these assumptions, Schurr's solution for the external potential of an insulating sphere is equivalent to that of a homogeneous sphere with a total current conductivity given by

$$\bar{\sigma}_c^* = \left( \lambda + \frac{\lambda_o \omega^2 \tau^2}{1 + \omega^2 \tau^2} \right) \frac{2}{a} - i \omega \left[ \epsilon'_c + \frac{2\lambda_o \tau}{a(1 + \omega^2 \tau^2)} \right], \quad (4)$$

where,  $\lambda = e_o \rho_d \mu_l$  and  $\lambda_o = e_o \rho_b \mu_l$ , are surface conductivities and  $\tau = a^2/2D_1$ , is the relaxation time of the charged particle or, equivalently, the time required to establish a stationary ionic distribution in its Stern layer under the action of a static field. Parameters  $\mu_l$ , and  $D_1$ , are the mobility and diffusion coefficient of the counterion in the charged layer,  $e_o$  the charge of the counterion and  $\epsilon'_c$  the real dielectric permittivity of the particle. Equation (4) reduces to the O'Konski result if  $\rho_b = 0$ , and to Schwarz's equation if  $\rho_d = 0$ .

According to equation (2), we may write from equation (4) that

$$\sigma_{c,ef} = \frac{2\lambda}{a} + \frac{2\lambda_o \omega^2 \tau^2}{a(1 + \omega^2 \tau^2)}; \quad (5a)$$

$$\epsilon_{c,ef} = \epsilon'_c + \frac{2\lambda_o \tau}{a(1 + \omega^2 \tau^2)}. \quad (5b)$$

Separating the real and imaginary components and using Kronig-Kramers' relation by which if  $\Re(\epsilon^*) = \epsilon'_c = \text{constant}$ , then  $\Im(\epsilon^*) = \epsilon''_c = 0$ , we obtain after some algebra,

$$\sigma_c^* = \frac{2(\lambda + \lambda_o)}{a} - \frac{2\lambda_o}{a(1 - i\omega\tau)}. \quad (6)$$

Thus, the dielectric response of a charged sphere and its associated Stern layer ionic atmosphere is uniquely described by a complex conductivity that exhibits a simple Debye relaxation with characteristic time  $\tau$  and limits  $\sigma_{c,0} = 2\lambda/a$  and  $\sigma_{c,\infty} = 2(\lambda + \lambda_o)/a$ . Thus, an Argand diagram of equation (6) is represented as a semicircle centered on the real axis, with intercepts at  $\sigma_{c,0}$  and  $\sigma_{c,\infty}$  on this axis.

The S-model polarization is a satisfactory approximation for cases where most counterions are tightly bound to the solid particles. However, neglecting the effect of ionic diffusion in the external double layer, produces a stronger enhancement in  $\sigma_{ef}$  and  $\epsilon_{ef}$  than that experimentally observed in clay suspensions (Arulanandan, 1969; Lockhart, 1980). To take into account this screening effect we have also considered a purely diffuse model for the ionic polarization of clays.

#### Diffuse layer polarization

To account for the motion of ions both tangentially and radially, Fixman (1980) developed an approximate theory based on the polarization of a diffuse Guoy-Chapman model. The thickness of the double layer was again assumed to be much smaller than the particle size. Fixman's formulation is based on a "far field" approximation for both the electro-

static and concentration potentials outside the diffuse layer and a pair of boundary conditions specified at its external limit. These boundary conditions were derived by integrating, over the thin diffuse layer, the difference between the exact ionic density distribution and the one resulting from the far field approximation. Next, he also required the tangential components of his diffusion or velocity potential function (defined as the sum of the electrostatic and the concentration potentials), to be continuous across the double layer. This results in the dielectric relaxation around a charged sphere being controlled by the diffusive fluxes of counterions within the double layer, but which extend in the bulk electrolyte beyond the limits of this layer. A similar observation has also been made by Dukhin and Shilov (1974) and Chew and Sen (1982) working on this problem using the same general theoretical approach.

With these approximations, Fixman's solution for the potential outside an insulating-charged sphere immersed in a symmetric monovalent electrolyte, can be written as

$$\Psi_w = \left[ -r - \frac{[1 - \delta_1(1 + Y)]a^3}{[2 + \delta_1(1 - 2Y)]r^2} \right] E_o \cos \theta, \quad (7)$$

with

$$Y = -\frac{1 + (1 - i)\sqrt{\omega\tau}}{2 + 2(1 - i)\sqrt{\omega\tau} - 2i\omega\tau}, \quad (8)$$

where  $\delta_1 = \beta_1/aC_1$ ,  $\beta_1$  being the surface counterion density necessary to balance the immobile charges on the particle and  $C_1$  the cationic concentration in the electrolyte far from the sphere. In equation (8),  $\tau$  has the same definition as in equation (4), representing the characteristic time required to build up a stationary charge configuration surrounding the sphere.

Equating equations (3) and (7), and using equation (2), we can show that

$$\bar{\sigma}_c^* = \frac{\delta_1 \bar{\sigma}_w^*}{1 - \delta_1 Y}, \quad (9)$$

or in terms of the effective properties

$$\sigma_{c,ef} = \frac{\delta_1(1 + \delta_1 A)\sigma_w - \omega\delta_1^2 B\epsilon_w}{(1 + \delta_1 A)^2 + \delta_1^2 B^2}, \quad (10a)$$

$$\epsilon_{c,ef} = \frac{\delta_1(1 + \delta_1 A)\epsilon_w + \delta_1^2 B\sigma_w/\omega}{(1 + \delta_1 A)^2 + \delta_1^2 B^2}, \quad (10b)$$

where  $A = (1 - y^3)/[2(1 - y^4)]$ ,  $B = y^2(1 - y)/[2(1 - y^4)]$  and  $y = \sqrt{\omega\tau}$ . Rearranging the components of equation (10a,b) using equation (2), we can write

$$\sigma_c^* = \frac{\delta_1 \sigma_w}{1 - \delta_1 Y}; \quad (11a)$$

$$\epsilon_c^* = \frac{\delta_1 \epsilon_w}{1 - \delta_1 Y}, \quad (11b)$$

where  $\sigma_w$  and  $\epsilon_w$  are the real parts of the conductivity and the permittivity of the aqueous electrolyte. Thus, in the case of a Guoy-Chapman ionic atmosphere, both the conductivity

and the dielectric permittivity of the system are represented as complex functions of frequency. From the behavior of  $Y$  for small and large  $\omega$  it can easily be shown that equation (11a) can be written as

$$\sigma_c^* = \sigma_{c,\infty} + \frac{\sigma_{c,0} - \sigma_{c,\infty}}{1 - 2i\omega\tau/(2 + \delta_1)[1 + (1 - i)\sqrt{\omega\tau}]}, \quad (12)$$

with the limits

$$\sigma_{c,0} = \frac{2\delta_1 \sigma_w}{2 + \delta_1}; \quad (13a)$$

$$\sigma_{c,\infty} = \delta_1 \sigma_w. \quad (13b)$$

Equation (12) has a slightly different dispersion pattern as compared with that of a Debye substance. It admits a Cole-Cole representation with the form (Cole and Cole, 1941)

$$\sigma_c^* = \delta_{c,\infty} + \frac{\sigma_{c,0} - \sigma_{c,\infty}}{1 + f(\omega, \tau)i^{-(1-\alpha)}} \quad (14)$$

where  $f(\omega, \tau) = 2\omega\tau/(2 + \delta_1)[(1 + \sqrt{\omega\tau})^2 + \omega\tau]$ . It is easy to show that for high  $\omega\tau$  values the complex ( $\sigma'$ ,  $\sigma''$ ) plane plot or Argand diagram of equation (14) tends to a skewed circular arc with its center located below the real axis. The angle between a radius through  $\sigma_{c,0}$  and the real axis is equal to 45 degrees ( $\alpha = 1/2$ ). For small  $\omega\tau$ , however, the plot resembles that of an ideal Debye substance.

Equation (13a) results in a wet-particle dc-conductivity that is a nonlinear function of cationic concentration  $C_1$  and hence of  $\sigma_w$ . From Einstein's relation,  $\sigma_w = \mu_e C_1 e_o$ , (where  $\mu_e$  is the cationic mobility and  $e_o$  the electronic charge) we may verify that  $\sigma_{c,0}$  increases with  $\sigma_w$  up to a constant value given by  $\sigma_{c,0} = \beta_1 \mu_e e_o/a$ . This behavior is consistent with the observed curvature (at dilute electrolyte concentrations) of experimental plots of the bulk conductivity of shaly sands as a function of the conductivity of the electrolyte (Johnson and Sen, 1988). Plots of  $\sigma_o$  versus  $\sigma_w$  generated using this model clearly show the nonlinear behavior discussed above (Lima and Sharma, 1991). Expressions similar to equations (12) and (13a, b) can also be written for the complex permittivity by simply replacing  $\sigma$  by  $\epsilon$  in these equations. The corresponding low- and high-frequency limits of  $\epsilon_{ef}$  are given by

$$\epsilon_{c,0} = \frac{2\delta_1}{2 + \delta_1} \left( \epsilon_w + \frac{\delta_1 \tau \sigma_w}{2 + \delta_1} \right); \quad (15a)$$

$$\epsilon_{c,\infty} = \delta_1 \epsilon_w, \quad (15b)$$

respectively. Thus, the observed electrolyte concentration dependence of the electrical properties of shaly sandstones can be accounted for with this model.

The boundary conditions introduced in the above models are accurate for values of  $\delta_1 < a/\kappa$ , where  $\kappa$  is the Debye screening distance ( $\kappa = [\epsilon_w k_B T / (2C_1 e_o^2)]^{1/2}$ , where  $k_B$  is the Boltzmann constant and  $T$  the absolute temperature). According to Sen (1987), such an approximation gives satisfactory results for  $\delta_1$  values below 16. We have restricted our calculations to this high limit.

## Extension to shaly materials

From the total current conductivity functions of a single particle we can derive the electrical properties of composite materials using generalized forms of Wagner and Bruggeman-Hanai theories (Hanai, 1968). Thus, for a dilute-disperse suspension of spherical particles we have from Wagner's theory

$$\bar{\sigma}_o^* = \bar{\sigma}_w^* \left[ \frac{2\bar{\sigma}_w^* + \bar{\sigma}_c^* - 2p(\bar{\sigma}_w^* - \bar{\sigma}_c^*)}{2\bar{\sigma}_w^* + \bar{\sigma}_c^* + p(\bar{\sigma}_w^* - \bar{\sigma}_c^*)} \right], \quad (16)$$

where  $\bar{\sigma}_o^* = \sigma_{o,ef} - i\omega\epsilon_{o,ef}$ ,  $\bar{\sigma}_c^* = \sigma_{c,ef} - i\omega\epsilon_{c,ef}$ , and  $\bar{\sigma}_w^* = \sigma_w - i\omega\epsilon_w$  are the total current conductivities of the suspension, individual particle and electrolyte, respectively. The volume fraction of solids in the mixture is  $p$ . It is important to note that our incorporation of the membrane polarization phenomena in the total current conductivity function ( $\bar{\sigma}_c^*$ ) allows us to use the above equation directly. This simplifies matters considerably. Equation (16) was derived under the assumption of a weak interaction between the spheres. So, it holds for small values of  $p$ . In practice, however, it has been found that equation (16) can be consistently applied even for fairly large  $p$ . After substitution of the above definitions and considering the limits of the effective properties at small and large  $\omega$ , equation (16) is rewritten as

$$\bar{\sigma}_o^* = \sigma_{o,0} - \frac{i\omega\tau_m[\sigma_{o,\infty} - \sigma_{o,0}]}{1 - i\omega\tau_m} - i\omega\epsilon_{o,\infty}, \quad (17)$$

where

$$\sigma_{o,0} = \sigma_w \left[ \frac{2\sigma_w + \sigma_{c,0} - 2p(\sigma_w - \sigma_{c,0})}{2\sigma_w + \sigma_{c,0} + p(\sigma_w - \sigma_{c,0})} \right]; \quad (18a)$$

$$\sigma_{o,\infty} = \epsilon_{o,\infty} \frac{\sigma_w}{\epsilon_w} - \frac{9p[\sigma_w\epsilon_{c,\infty} - \epsilon_w\sigma_{c,\infty}]}{[2\epsilon_w + \epsilon_{c,\infty} + p(\epsilon_w - \epsilon_{c,\infty})]^2}; \quad (18b)$$

$$\epsilon_{o,\infty} = \epsilon_w \left[ \frac{2\epsilon + \epsilon_{c,\infty} - 2p(\epsilon_w - \epsilon_{c,\infty})}{2\epsilon_w + \epsilon_{c,\infty} + p(\epsilon_w - \epsilon_{c,\infty})} \right]; \quad (18c)$$

and

$$\tau_m = \frac{2\epsilon_w + \epsilon_{c,ef} + p(\epsilon_w - \epsilon_{c,ef})}{2\sigma_w + \sigma_{c,ef} + p(\sigma_w - \sigma_{c,ef})}. \quad (18d)$$

Complex plane plots of the real versus imaginary part of equation (17) produce circular arcs with skewed spurs. Equation (18d) shows that the dielectric behavior of a charged suspension of spheres is characterized by a continuous distribution of relaxation times, since both  $\sigma_{c,ef}$  and  $\epsilon_{c,ef}$  are smooth functions of frequency. Although equations (16) and (17) are not appropriate to model precisely the dielectric response of shales (since they are only valid for dilute suspensions of solids), they are explicit expressions and are therefore useful to study the effects of the various governing factors on the behavior of clay-water mixtures. They are also helpful in designing representative circuit analogs for such materials.

For more concentrated suspensions the Bruggeman-Hanai equation (Hanai, 1968; Sen et al., 1981) is more appropriate,

$$\frac{\bar{\sigma}_o^* - \bar{\sigma}_c^*}{\bar{\sigma}_w^* - \bar{\sigma}_c^*} \left( \frac{\bar{\sigma}_w^*}{\bar{\sigma}_o^*} \right)^{1/3} = \phi, \quad (19)$$

where  $\phi$  is the volume fraction of the electrolyte or the porosity of the mixture. Although some doubts have been raised against the theoretical validity of Bruggeman's approach (Wobschall, 1977), equation (19) has been validated experimentally for concentrated mixtures of various kinds of materials (Hanai, 1968; Sen et al., 1981; Feng and Sen, 1985).

Equation (19) has the following frequency limits. When  $\omega \rightarrow 0$

$$\frac{\sigma_{o,0} - \sigma_{c,0}}{\sigma_w - \sigma_{c,0}} \left( \frac{\sigma_w}{\sigma_{o,0}} \right)^{1/3} = \phi, \quad (20a)$$

and

$$\begin{aligned} & \epsilon_{o,0} \left[ \frac{3}{\sigma_{o,0} - \sigma_{c,0}} - \frac{1}{\sigma_{o,0}} \right] \\ &= 3 \left[ \frac{\epsilon_w - \epsilon_{c,0}}{\sigma_w - \sigma_{c,0}} + \frac{\epsilon_{c,0}}{\sigma_{o,0} - \sigma_{c,0}} \right] - \frac{\epsilon_w}{\sigma_w}. \end{aligned} \quad (20b)$$

When  $\omega \rightarrow \infty$

$$\frac{\epsilon_{o,\infty} - \epsilon_{c,\infty}}{\epsilon_w - \epsilon_{c,\infty}} \left( \frac{\epsilon_w}{\epsilon_{o,\infty}} \right)^{1/3} = \phi, \quad (21a)$$

$$\begin{aligned} & \sigma_{o,\infty} \left[ \frac{3}{\epsilon_{o,\infty}} - \epsilon_{o,\infty} - \frac{1}{\epsilon_{o,\infty}} \right] \\ &= 3 \left[ \frac{\sigma_w - \sigma_{c,\infty}}{\epsilon_w - \epsilon_{c,\infty}} + \frac{\sigma_{c,\infty}}{\epsilon_{o,\infty} - \epsilon_{c,\infty}} \right] - \frac{\sigma_w}{\epsilon_w}. \end{aligned} \quad (21b)$$

In general, equation (19) cannot be explicitly written in a simple closed form, in terms of the high- and low-frequency limits of the effective electrical parameters of the mixture. Therefore, it was solved numerically for specific combinations of the component properties, using the Laguerre method of finding the roots of polynomials with complex coefficients. One of such complex roots, yields the effective properties of the composite material. We used a Fortran subroutine implemented by Press et al. (1986). The ambiguity in identifying the correct solutions was removed by considering the functional behavior of the solutions at the high and low frequency limits.

Equations (16) and (19) can also be used to describe electrolytic suspensions of spherical sand grains coated by an aggregate of wet clays. For this,  $\bar{\sigma}_c^*$  has to be replaced by an effective total current conductivity ( $\bar{\sigma}_{CS}^*$ ) for the uniformly coated sphere. To compute the effective total current conductivity of a clay-coated sphere we compare the external electric potential around the layered sphere immersed in a homogeneous electrolyte of total current conductivity  $\bar{\sigma}_w^*$  and under the influence of a uniform electric field, with that of a single equivalent homogeneous sphere under the same conditions (Lima and Sharma, 1990). Thus, we get

$$\bar{\sigma}_{CS}^* = \bar{\sigma}_c^* \left[ \frac{2p\bar{\sigma}_c^* + (3 - 2p)\bar{\sigma}_a^*}{(3 - p)\bar{\sigma}_c^* + p\bar{\sigma}_a^*} \right], \quad (22)$$

where  $\bar{\sigma}_c^*$  and  $\bar{\sigma}_a^*$  are the total current conductivities of the coating clay aggregate and spherical nucleus, respectively. In equation (22)  $p = 1 - (a/b)^3$ , is the volume fraction of the clay shell, with  $a$  and  $b$  as the radius of the core and of the composite sphere. Thus, by assuming the rock grains as spheres of arbitrary sizes coated with clays, we can compose a self-similar mixture whose electrical properties are described by equation (19). Such an approach has been formally validated by Sen et al. (1981) in a model for the electrical response of clean wet sandstones.

Equivalent expressions for a dispersed model where the clays occur as discrete particles in a sand matrix have to rely on the more complicated three-component effective medium formulation (Feng and Sen, 1985; Lima and Sharma, 1990) and will not be discussed here.

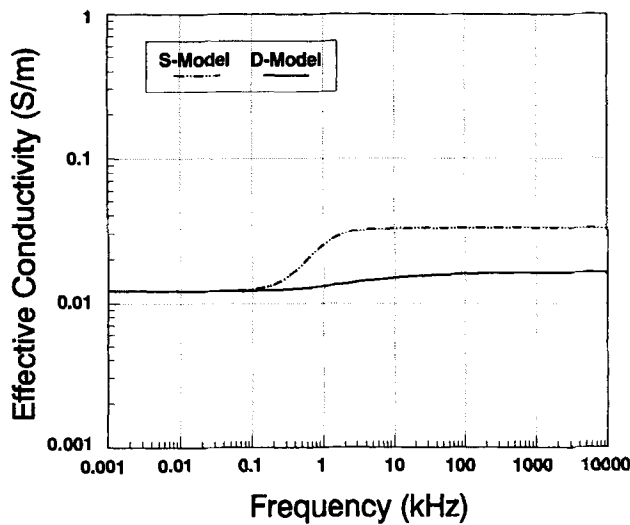
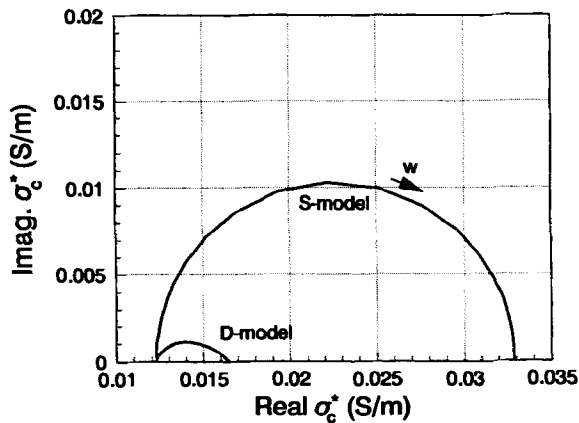


FIG. 1. The dependence of the effective conductivity on frequency for a single-charged wet sphere. Parameters:  $a = 9.0 \times 10^{-7}$  m;  $\beta_a = 1.2 \times 10^{18}$  /m<sup>2</sup>;  $\sigma_w = 0.024$  S/m;  $\epsilon_w = 6.9 \times 10^{-10}$  F/m;  $\epsilon_c = 3.1 \times 10^{-11}$  F/m;  $D_1 = 2.0 \times 10^{-9}$  m<sup>2</sup>/s.



a)

## NUMERICAL AND EXPERIMENTAL RESULTS

In Figures 1 and 2 we compare theoretical results obtained with equations (5a, b) and (10a, b) for the effective properties of a single charged sphere, using data typical of kaolinite clays as summarized in Figure 1. We have assumed a  $\rho_b$  value for the Stern layer model (S-model) such that in the low-frequency range, both models show the same  $\sigma_{c,ef}$ . This particular situation required about 63 percent of the compensating counterions to be stored in the fixed part of the double layer. The figures show that the diffuse model (D-model) has a lower amplitude of variation in its effective parameters, as compared with the S-model, and also has a broader dispersion spectrum, with a characteristic frequency in the kilocycle range.

The Argand diagrams of Figure 3a, b show the complex conductivities of a single sphere according to equations (6a) and (12). The S-model plots as a semicircle centered on the

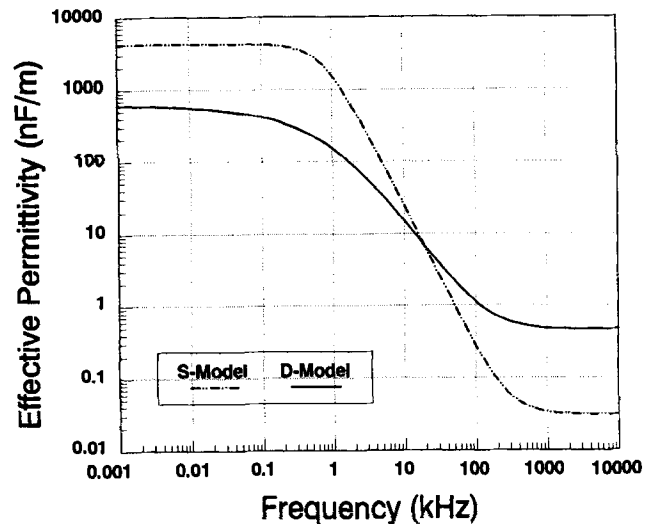
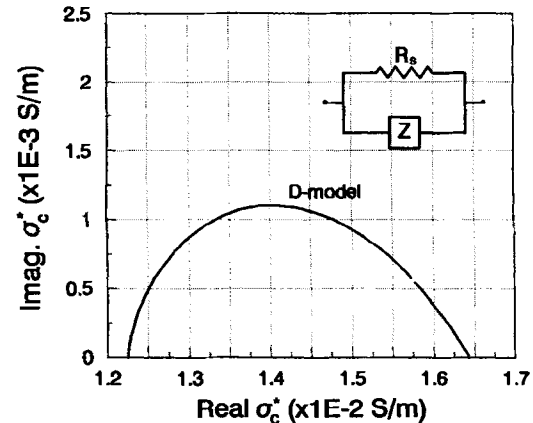


FIG. 2. Dependence of effective permittivity on frequency for a single-charged wet sphere. Parameters as in Figure 1.



b)

FIG. 3. Argand diagrams for the complex conductivity of a single-charged wet sphere. Parameters as in Figure 1.

real axis. The D-model diagram, on the other hand, is depicted as an asymmetric curve with its high-frequency portion being represented by a depressed circular arc, as discussed before. The enhancement depicted on the S-model curve is due to an increase in the imaginary component of its electrical conductivity, as a result of neglecting the ionic diffusion in the external part of the double layer (Fixman, 1980; Lyklema et al., 1983). The equivalent circuit inserted in Figure 3b simulates the complex conductivity of the D-model. The resistor  $R_s$ , proportional to  $\sigma_{e,\infty}$ , corresponds to a steady ionic transport through the double layer that is active even at high frequencies. The complex impedance  $Z$ , given by the second term on the right side of equation (12), constitutes a parallel ionic path that is more effective at low frequencies. It provides both an in-phase current due to the applied electric field and an out-of-phase current due to charge polarization and ionic concentration gradients. Figure 4 contains equivalent representation of the same basic data in terms of the total current conductivity functions given by equations (4) and (9). The vertical straight segments in these diagrams allows us to introduce in the circuit analog, the additional capacitance (proportional to  $\epsilon_{e,\infty}$ ) connected in parallel with  $R_s$  as depicted in Figure 7.

Figures 5 and 6 show the modeling of clay-water mixtures treated as dilute-disperse suspensions in accordance with equation (16). A pattern of amplitude variations with frequency similar to that for a single sphere is observed. The S-model curves show the existence of two relaxation times: one due to diffusion in the fixed counterion layer and another resulting from the permittivity contrast between the media (the conventional Maxwell-Wagner effect). The D-model, however, as for the isolated particle case, has a wide spectrum simulating a continuous distribution of relaxation times between these two extremes. These figures also contain, for comparison purposes, experimental data for a sodium kaolinite suspension with 36.5 percent and 39 percent of clay by weight, as reported by Arulanandan (1969) and by Lockhart (1980). Although the individual clays are far from being spherical, we find good agreement with the D-model simulation using a clay parameter  $\beta_1/a = 1.333 \times 10^{24} \text{ m}^3$ . A water conductivity  $\sigma_w = 1.2 \times 10^{-3} \text{ S/m}$  was used in both our simulations and their experiments. This

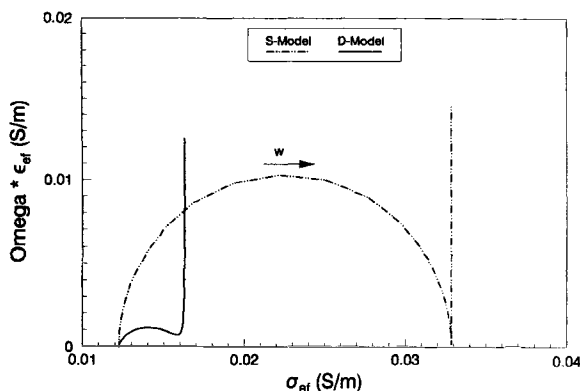


FIG. 4. Complex plane plots of the total current conductivity functions for the single charged sphere. Parameters as in Figure 1.

gives a characteristic relaxation time  $\tau \approx 2.0 \times 10^{-4} \text{ s}$ . Numerical results for the same input data were also obtained using the Bruggeman-Hanai equation (19). The results were for all practical purposes indistinguishable from those plotted in Figures 5 and 6 using equation 16.

The data from the total current conductivity functions (equation 17) of these suspensions were also plotted as Argand diagrams in the complex plane (Figure 7). The curves are represented as depressed circular arcs with spurs at the high-frequency limit, similar to the ones experimentally observed in nonideal admittance plots of dielectric solids (Jonscher, 1975). A maximum in the imaginary part of equation (17), located within the frequency range of each circular arc, occurs at  $\omega \approx 1/2\pi\tau'_m$ , with  $\tau'_m$  being a mean characteristic time for the relaxation of that suspension. In

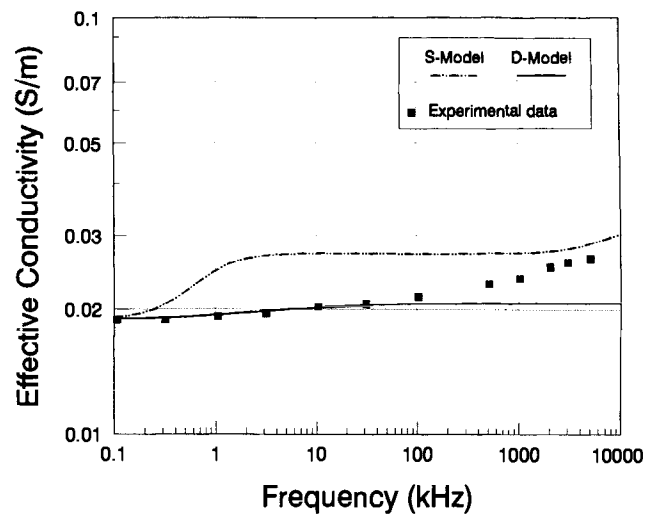


FIG. 5. Conductivity dispersion of a dilute suspension of charged spheres. Comparison with experimental data for a kaolinite gel with 39 percent of clays according to Arulanandan (1969).  $\beta_1/a = 1.333 \times 10^{24} \text{ m}^3$ ,  $p = 0.39$ .

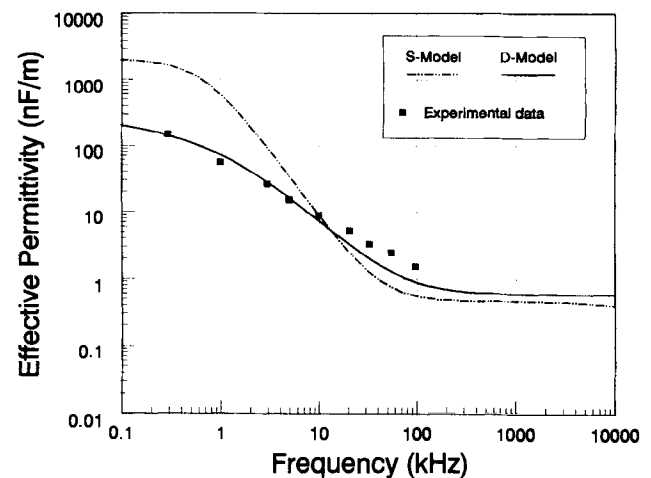


FIG. 6. Permittivity dispersion of a dilute suspension of charged spheres. Comparison with experimental data of a kaolinite gel with 36.7 percent of clay according to Lockhart (1980).  $\beta_1/a = 1.333 \times 10^{24} \text{ m}^3$ ,  $p = 0.39$ .

the high-frequency limits, the capacitive effect represented by  $\omega\epsilon_{o,\infty}$  gives rise to the ascending spurs in the diagrams. These features are duplicated by the circuit analog depicted in Figure 7b. This circuit is quite similar to the one used by Dias (1972) to fit experimental data on the membrane polarization of rocks. It also corresponds, in essence, to the equivalent circuit used by Knight (1984) for modeling the dielectric response of sandstones. The present analysis of course provides us with an explicit physical and mathematical representation for each component in the equivalent circuit in terms of basic model parameters.

The Bruggeman-Hanai equation (19) was also used to model typical shale data as displayed in Figure 8. The data was reported by Scott et al. (1967) for the Mancos shale, an Early Cretaceous dark gray carbonaceous shale from eastern Utah. The experimental values were obtained for a distilled water content of 3.8 percent by volume. We assume this shale to be a compacted clay-water mixture having the following component properties: water conductivity  $\sigma_w = 2 \times 10^{-2}$  S/m and clay parameter  $\beta_1/a = 6.0 \times 10^{23}/\text{m}^3$  (which is a typical value for an illite clay). For these input data, we predict a frequency dependence of the effective conductivity and permittivity that is remarkably close to the measured curves, using a D-model.

Theoretical results using equations (19) and (22) for shales and sandstones with a clay coating structure are shown in Figures 9 to 11. Figure 9 illustrates the effects of changing the clay parameter  $\beta_1/a$  on the electrical dispersion of idealized shales. It is evident that as  $\beta_1/a$  increases the characteristic time for double layer relaxation ( $\tau$ ) decreases and the frequency at which large changes in dielectric constant occur correspondingly increases.

The predicted variation in the effective properties of a shaly sandstone with 20 percent of porosity, as a function of frequency and clay proportion in the solid matrix, is shown in Figure 10. These overall trends are in qualitative agree-

ment with many experimental measurements described in the literature (Mehran and Arulanandan, 1977; De Loor, 1983; Kenyon, 1984).

In Figure 11 we show the frequency behavior of the dielectric parameters of a shaly sandstone also depending on the saturating electrolyte conductivity. These simulations reveal that the magnitude of the bulk effective conductivity increases with the electrolyte concentration in conformity with many experimental observations. The effective permittivity, however, may behave as a decreasing function of  $\sigma_w$ . Although an adequate set of dielectric permittivity data was not available to us, some data reported by Vinegar and Waxman (1984) for a shaly sand equivalent quadrature conductivity reflect this same behavior. However, some sandstone measurements in the frequency range from 1 to  $10^4$  MHz (Kenyon, 1983) show that, in fact,  $\epsilon_{o,ef}$  increases with  $\sigma_w$ . This behavior can also be reproduced by the present approach, assuming the water salinity inside the shale coating is almost constant and independent of the free pore electrolyte (Lima and Sharma, 1991).

Table 1 provides a potential method for the application of the theory to wireline measurements. Obtaining the six reservoir parameters listed from the six wireline measurements will require inverting the equations in this paper. Since the equations are coupled and nonlinear this will be a nontrivial task. This would be a fruitful avenue for future research in applying the theory presented.

## CONCLUSIONS

Two simplified approaches have been considered to simulate the frequency-dependent electrical properties of materials containing charged particles such as clays in soils and rocks. The clays were assumed to consist of an insulating-charged nucleus surrounded by a thin layer of electrolyte containing a double layer counterion atmosphere. For the

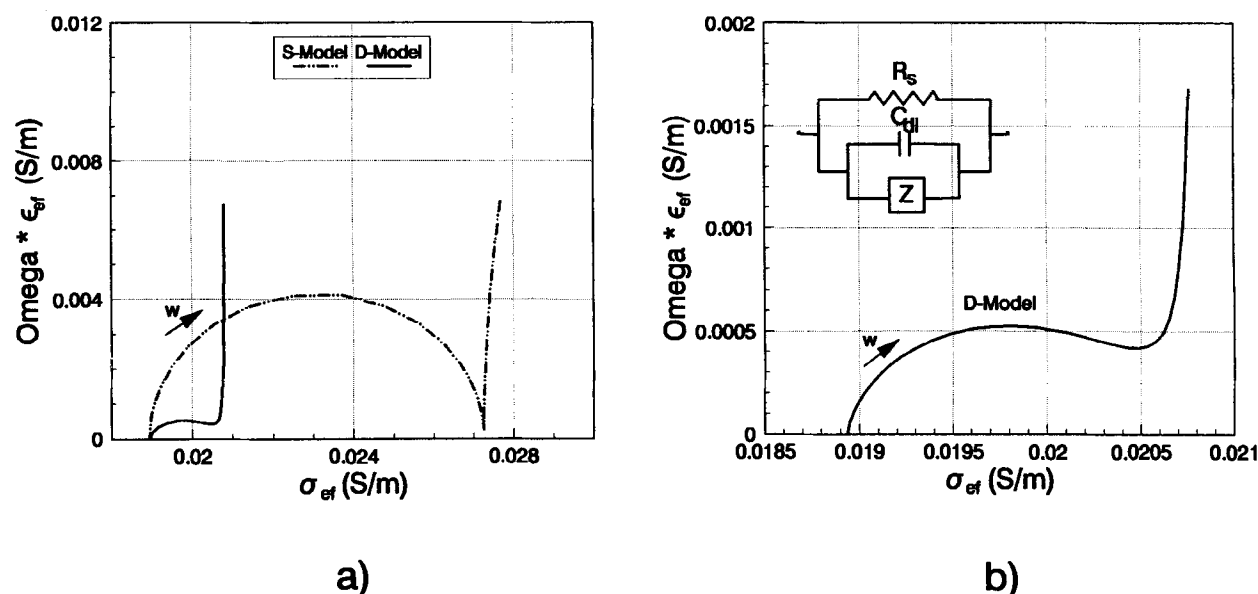


FIG. 7. Argand diagram for the total current conductivity functions of dilute suspensions of charged spheres. Parameters as in Figures 6 and 7.

Stern layer (S-model), we use the results from Schurr (1964), whereas for the diffuse layer D-model we use Fixman's (1980) model as a starting point. The effective conductivity and permittivity functions of the grains and their associated double layers were derived. These effective properties allow us to use generalized mixture theories to obtain the behavior of shaly sandstones.

A macroscopic representation in terms of a total (volume) current conductivity for a charged particle was shown to be a sound approach to describe the dielectric dispersion of shaly sands. From the total current conductivity of wet particles derived for both models we found that the electrical properties of heterogeneous mixtures can be characterized as complex functions of frequency by applying general mixture theories.

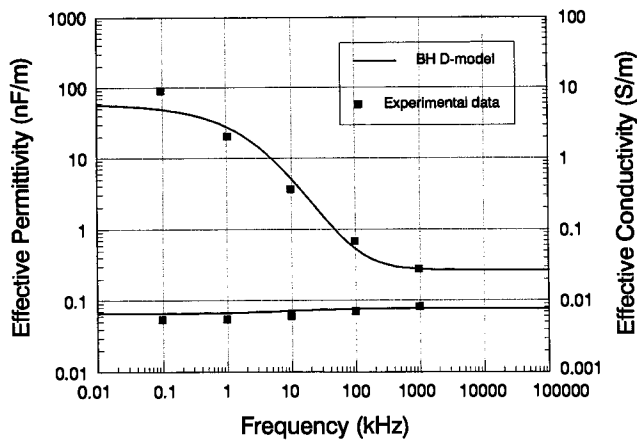


FIG. 8. Modeling the effective conductivity and permittivity of the Mancos shale. Experimental data according to Scott et al., (1967). Parameters:  $\beta_1/a = 6.0 \times 10^{23} \text{ /m}^3$ ;  $\sigma_w = 0.02 \text{ S/m}$ ;  $\epsilon_w = 6.9 \times 10^{-10} \text{ F/m}$ ;  $D_1 = 2.0 \times 10^{-9} \text{ m}^2/\text{s}$ ;  $\phi = 4 \text{ percent}$ .

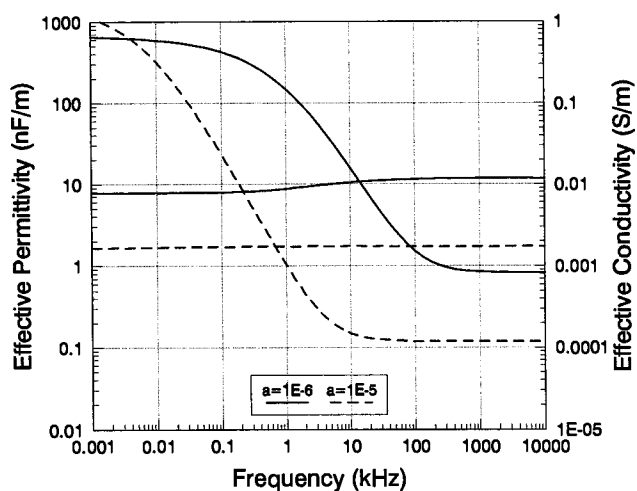


FIG. 9. Effective conductivity and permittivity of an idealized shale of porosity  $\phi = 10 \text{ percent}$  for two different clay volume fractions  $p$ . Further parameters:  $\sigma_w = 0.01 \text{ S/m}$ ;  $\epsilon_w = 6.9 \times 10^{-10} \text{ F/m}$ ;  $D_1 = 2.0 \times 10^{-9} \text{ m}^2/\text{s}$ .

On the basis of this approach, we have expressed the effective electrical parameters of a shaly sand in terms of the geometrical and physical properties of its components. These properties include the charge density and grain size of the clays, and the conductivity and permittivity of the electrolyte. The D-model with its wider dispersion appears to be more appropriate for describing the electrical response of clay-water suspensions, shales and shaly sandstones in the frequency range from 1 to  $10^7 \text{ Hz}$ . The zero frequency limit of their conductivity functions are nonlinear functions of the electrolyte salinity at low concentrations, in conformity with experimental evidence.

Although the assumed spherical geometry does not represent the real shape of mineral grains in sedimentary rocks, the results obtained with this representation describe fairly

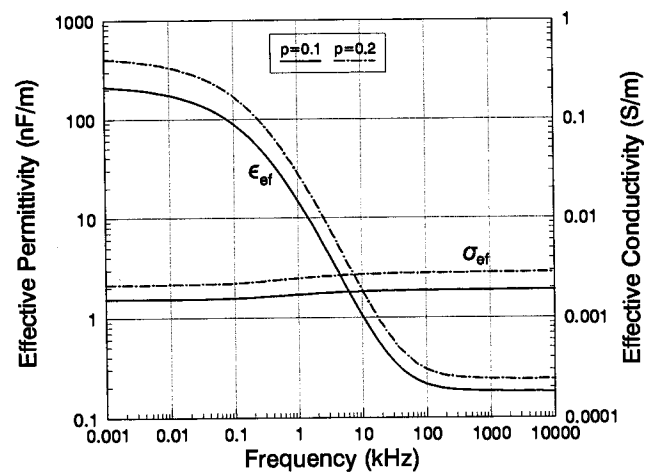


FIG. 10. Effective properties of idealized shaly sandstones of porosity  $\phi = 20 \text{ percent}$ , for two different clay volume fractions  $p$ .  $\beta_1/a = 1.0 \times 10^{24} \text{ /m}^3$ . Other parameters as in Figure 9.

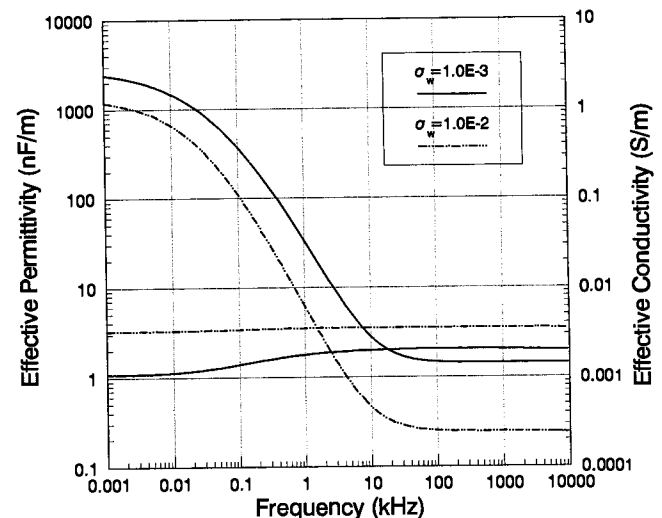


FIG. 11. Effective properties of a shaly sandstone of porosity  $\phi = 20 \text{ percent}$  and  $p = 10 \text{ percent}$  for two different electrolyte conductivities  $\sigma_w$ . Parameters as in Figure 10.

Table 1. Potential application of the theory.

| 6 Measurements                                                                                                                                                                                                                                                               | 6 Unknowns                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Low Frequency (&lt; 100 Hz)</p> <p><math>\sigma_o</math> Flushed zone</p> <p><math>\sigma_o</math> Virgin formation</p> <p>High Frequency (&gt; 1MHz)</p> <p><math>\sigma_o, \epsilon_o</math> Flushed zone</p> <p><math>\sigma_o, \epsilon_o</math> Virgin formation</p> | <p><math>\Phi</math> (Porosity)</p> <p><math>S_{xo}</math> (Water saturation)</p> <p><math>S_w</math> (Water saturation)</p> <p><math>p</math> (Clay fraction)</p> <p><math>B/a</math> (Clay parameter)</p> <p><math>\alpha_w</math> (Water resistivity)</p> |

well the essential features of the dispersion spectra of the electrical properties of shaly materials and other charged systems. The results are also useful in simulating the electrical behavior of rocks in terms of ideal circuit elements such as resistors and capacitors combined in parallel with a frequency-dependent complex element that is similar to a diffusion controlled Warburg impedance. Finally, on the basis of the equations presented here, it is possible to solve for shaly sandstone parameters using the multifrequency electromagnetic data measured in boreholes with the available electric logging tools. Additional experimental data is needed to validate the model. In addition, the model equations need to be inverted so that reservoir parameters can be estimated from measurements of conductivity and dielectric constant at various frequencies.

#### ACKNOWLEDGMENT

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